3. Two cylindrical rods of same length have the area of cross section in the ratio 2:1. If both the rods are made up of same material, which of them conduct heat faster?

   a) Both rods b) Rod-2

   c) Rod-1 d) None of them

4. In which mode of transfer of heat, molecules pass on heat energy to neighbouring molecules without actually moving from their positions?

   a) Radiation b) Conduction

   c) Convection d) Both B and C

5. A device in which the loss of heat due to conduction, convection and radiation is minimized is

   a) solar cell b) solar cooker

   c) thermometer d) thermos flask

II. Fill in the blanks.

1. The fastest mode of heat transfer is _____.
2. During day time, air blows from _____ to _____.
3. Liquids and gases are generally _____ conductors of heat.
4. The fixed temperature at which matter changes state from solid to liquid is called _____.

III. Assertion and reason type questions.

Mark the correct choice as:

a. If both assertion and reason are true and reason is the correct explanation of assertion.

b. If both assertion and reason are true but reason is not the correct explanation of assertion.

c. If assertion is true but reason is false.

d. If assertion is false but reason is true.

1. **Assertion:** Food can be cooked faster in vessels with copper bottom.

   Reason: Copper is the best conductor of heat.

2. **Assertion:** Maximum sunlight reaches earth's surface during the noon time.

   Reason: Heat from the sun reaches earth's surface by radiation.

3. **Assertion:** When water is heated up to 100°C, there is no raise in temperature until all water gets converted into water vapour.

   Reason: Boiling point of water is 10°C.

IV. Answer briefly.

1. Define conduction.
2. Ice is kept in a double-walled container. Why?
3. How does the water kept in an earthen pot remain cool?
4. Differentiate convection and radiation.
5. Why do people prefer wearing white clothes during summer?
6. What is specific heat capacity?
7. Define thermal capacity.
8. Define specific latent heat capacity.

V. Answer in detail.

1. Explain convection in daily life.
2. What are the changes of state in water? Explain.
3. How can you experimentally prove water is a bad conductor of heat? How is it possible to heat water easily while cooking?

VI. Numerical Problems.

1. What is the heat in joules required to raise the temperature of 25 grams of water from 0 °C to 100 °C? What is the heat in Calories? (Specific heat of water = 4.18 J/g °C) **(Ans. 10450 J)**
2. What could be the final temperature of a mixture of 100 g of water at 90 °C and 600g of water at 20°C. **(Ans. 30°C)**
3. How much heat energy is required to change 2 kg of ice at 0°C into water at 20°C? (Specific latent heat of fusion of water = 3,34,000 J/kg, Specific heat capacity of water = 4200 $JKg^{-1}K^{-1}$). **(Ans. 8,36,000 J)**

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